SOURAV TIWARI

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LinkedIn | GitHub | Portfolio

CAREER OBJECTIVE

B.Tech Computer Science & Information Technology student graduating in 2027 with hands-on experience in software development, backend engineering, and full-stack applications. Proficient in C++, Java, Python, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, REST APIs, and SQL. Experienced in designing, developing, testing, debugging, and optimizing software solutions through internships and projects, with experience working in Agile development environments.

TECHNICAL SKILLS

Languages:	Java, C, C++, Python, SQL
Computer Science:	Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering, Complexity Analysis
Development:	HTML5, CSS3, JavaScript (ES6+), DOM Manipulation, JSON, AJAX, Responsive Web Design, REST APIs, Node.js, Express.js
Databases:	MySQL, MongoDB, Database Design, Query Optimization
Tools & Platforms:	Git, GitHub, Linux, Postman, AWS, Google Cloud
ML & AI:	Machine Learning, LLM Integration, Ollama, Prompt Engineering, AI Agents, Google Gemini AI

INTERNSHIPS

AI/ML and Software Development Intern – Government Of India Organisation

Stipend: ₹30,000/month *Sept 2026 – Present*

- Contributing to software development, debugging, testing, and backend-oriented technical tasks.
- Working with development tools, databases, APIs, and secure software engineering practices.

Stack: *React.js, Python,AI/ML, PostgreSQL, Ubuntu, Linux, Fast API, Docker*

ML and Backend Developer Intern — *Coding Arena*

Mar 2026 – Jun 2026

- Designed and implemented **15+ REST API endpoints** using Node.js and Express.js, reducing average API response time by **25%** through backend optimization.
- Developed full-stack modules integrating React.js interfaces with Node.js services and designed database schemas, contributing to **3 Agile development sprints**.
- Developed and optimized machine learning models in Python, improving prediction accuracy by 20% across 5 classification pipelines while automating data workflows to save 8 hours per week.

Stack: *Node.js, Express.js, React.js, Python, Machine Learning, REST APIs, MongoDB, Agile*

PROJECTS

Hospital Management System — *Anveshana 2025*

- Designed and developed a full-stack hospital management application using React, Node.js, Express.js, and MongoDB for managing **100+ patient, doctor, and room records**.
- Implemented **RESTful APIs, JWT authentication, CRUD operations, and real-time data updates**, improving data access efficiency by approximately **30%**.
- Documented system architecture, API specifications, and test cases to improve maintainability and software quality.

Stack: *Node.js, Express.js, React, MongoDB, TypeScript, JWT Authentication, RESTful API*

AI/ML Crypto Primitive Identification in Firmware — *SIH 2025*

- Built an AI/ML pipeline to identify cryptographic primitives in firmware binaries, performing root-cause analysis of false positives and defining validation scenarios to improve detection accuracy and reduce manual analysis effort by ~40%.
- Analyzed software-hardware behavior of firmware modules and documented system specifications, performance requirements, and validation results for stakeholder review.

Stack: *Python, PyTorch, NumPy, NetworkX, Flask, HTML, CSS, JavaScript*

EDUCATION

Bachelor of Technology — Computer Science & Information Technology
Sagar Institute of Research and Technology, Bhopal

CPI: 8.20/10.00 | Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks *2023 – 2027*

Intermediate (PCM), Government Model Co-Education Senior Secondary School, Punjab / *88.4%* *2021 – 2023*
Matriculation, St. Pius Convent Senior Secondary School, Punjab / *96.99%* *2019 – 2021*

CERTIFICATIONS

- NPTEL Introduction to Machine Learning — NPTEL | Scored 57% | 2026
- Google Cloud Arcade — Google | Completed 10+ cloud labs, multiple GCP skill badges | June 2025
- NPTEL Programming in Java — NPTEL | Scored 68% | 2025

ROLES & RESPONSIBILITIES

- Microsoft Learn Student Ambassador | Microsoft | April 2025 – Present — Engaged 100+ learners in web development and cloud topics across 5+ workshops.

EXTRA & CO-CURRICULAR ACTIVITIES

- Publications: Research paper published and presented at an International Conference (IJAR, ISSN: 2455-6211), June 2024.
- SIH 2025 Finalist — Selected among top teams nationally out of 10,000+ Smart India Hackathon entries.
- YUKTI 2025 & Anveshana 2025 — Reached the semifinals in two national-level innovation competitions.
- SolveExpo 2026 — Awarded 3rd Runner-Up for presenting an innovative web-based project to 50+ judges.